

PHYSICS 7B

UC BERKELEY

FALL 2020

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## 7B Discussion Problems, Week 9

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*Topics*

Capacitance, Electric Potential, and Electric Fields

March 16, 2026

## Week 9 Discussion Problems

1. Suppose one plate of a parallel-plate capacitor is tilted so it makes a small angle  $\theta$  with the other plate, as shown below. Determine a formula for the capacitance  $C$  in terms of  $A$ ,  $d$ , and  $\theta$ , where  $A$  is the area of each plate. Assume the plates are square. Hint: consider the capacitor as many capacitors in parallel.

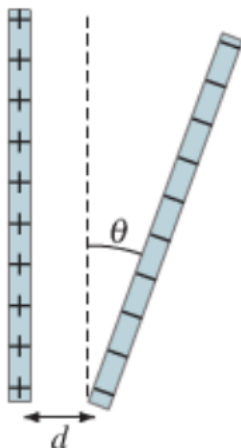
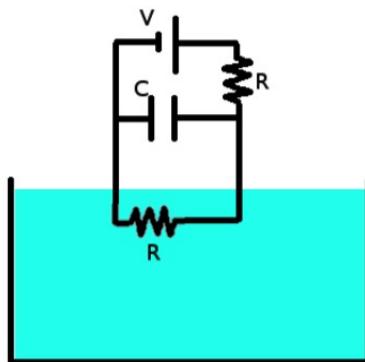


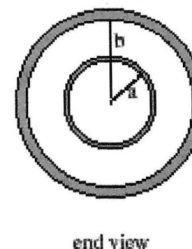
Figure 1: Capacitor with one plate tilted relative to the other.

2. An initially uncharged capacitor with capacitance  $C$  is connected to two resistors of resistance  $R$  and a battery with voltage  $V$  in the arrangement below: The resistor in parallel to the capacitor is immersed



in a vat of water. There is a mass  $m$  of water (with specific heat  $c_w$ ) initially at temperature  $T_0$ . Assume  $m$  is large enough and  $T_0$  far enough from the boiling/freezing point so no phase change occurs. What is the temperature of the water as a function of time? What assumptions did you have to make?

3. In this problem we will calculate the capacitance of a pair of concentric metal cylinders.
  - (a) Imagine placing charges  $\pm q$  on the plates, and sketch the resulting electric field in the region between the cylinders. (Say, the inner cylinder has  $+q$  and the outer one has  $-q$ .)
  - (b) Calculate the strength of the electric field at points between the cylinders.



- (c) Find the difference in potential between the plates, by integrating the field you found in part (b) from one plate to the other.
  - (d) What is the capacitance of the pair of cylinders?
4. Consider a solid sphere surrounded by a thin spherical shell. The inner sphere has radius  $a$  and total charge  $-Q$  distributed throughout its volume; the outer shell has radius  $b$  and total charge  $+Q$ .
  - (a) Determine the capacitance of this configuration
  - (b) What happens in the limit where  $b \rightarrow \infty$ ? What system does this describe?
5. What does it mean when we say the electric field is “conservative”?
6. (Birgeneau Midterm 2018 Problem 8) Consider an infinite plane of uniform surface charge density  $\sigma$  in the  $x$ - $y$  plane that has a circular hole of radius  $R$  cut out from it.
  - (a) Calculate the electric field a distance  $z$  above the center of the hole
  - (b) Determine the field in the limit  $z \gg R$ . Explain the physical interpretation of the result.
  - (c) Determine the field in the limit  $z \ll R$ . Explain the physical interpretation of the result.
7. A very long line of charge with positive linear charge density  $\lambda$  lies along the  $x$ -axis. An identical line of charge lies along the  $y$ -axis.
  - (a) Determine the electric field  $E(x, y)$  for all points in the  $x$ - $y$  plane.
  - (b) Determine the change in electrostatic potential  $\Delta V$  between the points  $(x = a, y = a)$  and  $(x = a, y = 3a)$ .
  - (c) How much work must be done to move a small negative charge  $-q$  from the point  $(x = 3a, y = 3a)$  to  $(x = a, y = a)$ ?
  - (d) For a very long linear charge distribution we do not define the zero of electrostatic potential to be at infinity. Why not?

